Final

ENV S 193 DS

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Setup chunk:

```
# installing all packages
library(tidyverse)
library(dplyr)
library(janitor)
library(here)
library(ggeffects)
library("DHARMa")
library(ggeffects)
```

Problem 1

1a.

He most likely used a logistic regression because they were testing whether the distant to water edge predicted for the probability of microhabitat use. In the second part, he most likely perform a chi square test of independence because they were testing whether bird species and habitat (which are categorical variables) were related.

1b.

Part A:

The American Avocets were more or less likely to use a microhabitat depending on their distance from the water. Distance to water's edge is a significant factor that influences microhabitat use.

Part B:

Use of habitat showed to be different across Avocets, Forster's Terns, and Black-necked Stilts. The three species showed unique patterns in the use of managed ponds, marshes, salt ponds, and mudflats.

1c.

An additional variable that could contribute to the probability of Avocet microhabitat use is human activity. Maybe it could be measured as a categorical variable organized in low, medium, and high levels, assessed by how many people are present in the habitat hourly. The birds may avoid those areas with more human activity because it might affect how much the birds go out of their shelters/nests to go eat, or may trigger them to fly out of their habitat when people are really close to them.

Problem 2

2a.

```
nests <- read_csv(here("data", "nest_data_final.csv"))
```

```
# Clean dataset
nests_clean <- nests |>
  # Select relevant columns according to metadata
     select(ant_species, height_cm, alive) |>
  # Select the relevant ant species
filter(ant_species %in% c("AB", "RRB", "BBR")) |>
mutate(
    alive = case_when(
      alive == "A" ~ 1,
      alive == "D" ~ 0
    ),
  # Reorder ant species
  # Set desired order
    ant_species = factor(ant_species,
                         levels = c("AB", "RRB", "BBR")),
    # Creating new columns displaying ant names
```

```r
    scientific_name = case_when(
      ant_species == "AB"  ~ "C. sjostedti",
      ant_species == "RRB" ~ "C. mimosae",
      ant_species == "BBR" ~ "C.nigriceps"))
```

```r
# Convert data into string and display structure
str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ ant_species    : Factor w/ 3 levels "AB","RRB","BBR": 2 2 2 2 1 2 3 2 1 3 ...
 $ height_cm      : num [1:270] 392 310 513 154 324 ...
 $ alive          : num [1:270] 1 1 1 1 1 1 1 1 1 1 ...
 $ scientific_name: chr [1:270] "C. mimosae" "C. mimosae" "C. mimosae" "C. mimosae" ...
```

```r
# Take 10 random samples
slice_sample(nests_clean, n = 10)
```

```
# A tibble: 10 x 4
   ant_species height_cm alive scientific_name
   <fct>           <dbl> <dbl> <chr>
 1 BBR              170      1 C.nigriceps
 2 RRB              201      1 C. mimosae
 3 RRB              300      1 C. mimosae
 4 RRB              234      1 C. mimosae
 5 RRB              341      1 C. mimosae
 6 RRB              279      1 C. mimosae
 7 RRB              174.     1 C. mimosae
 8 RRB              116.     1 C. mimosae
 9 RRB              189      1 C. mimosae
10 RRB              236.     1 C. mimosae
```

2b.

In the case column, a value of 1 means that a bird nest was present in the tree habitat while 0 indicates the absence of a bird nest. The values 1 and 0 represent if a tree was used as a nesting site or not.

2c.

The predictor variables are tree height and acacia ant species.

Tree height is a continuous quantitative variable, because it can be measured. As the height increases the probability of more nests appearing is expected to increase because the trees get bigger and taller, providing more spaces for birds to form shelteres that are not in the proximity of any predators that could probably harm them. This information is supported in the paper's results section where the probability of a bird deciding to nest increased when the trees increased by a 1 meter interval.

Ant species is a categorical variable, because it has unique categories of something, in this case, there are distinctive, unique species of ants. Ant species may also influence the probability of nest appearance, and the abstract and introduction of Lujan's paper describes the relationship of ants and birds to be mutualistic.

2d.

Table Only:

```
# Make models fit logistic regression model with response and predictor variables
model1 <- glm(
 alive ~ height_cm + ant_species,
  data = nests_clean,
  family = binomial
)
```

```
# Make models fit log. regression model w/ response + predicting variables
model2 <- glm(
alive ~ height_cm * ant_species,
  data = nests_clean,
  family = binomial)
```

2e.

Code only:

```
# Create table
table <- tibble(
  # Create models
  model_number = c(1, 2),
  # Differentiate them by interaction terms
  model_description = c(
    "Nest occurrence predicted by height and species without interactions",
    "Nest occurrence predicted by height species with interaction btwn variables."
  ),
  # Include predictor variables
  predictors = c(
    "height_cm + ant_species",
    "height_cm * ant_species"
```

```
  ),
  # Indicate presence of interactive term
  interaction_term = c("No", "Yes")
)

# Use knitr function to embed table into quarto
knitr::kable(table)
```

model_number	model_description	predictors	interaction_term
1	Nest occurrence predicted by height and species without interactions	height_cm + ant_species	No
2	Nest occurrence predicted by height species with interaction btwn variables.	height_cm * ant_species	Yes

2f.

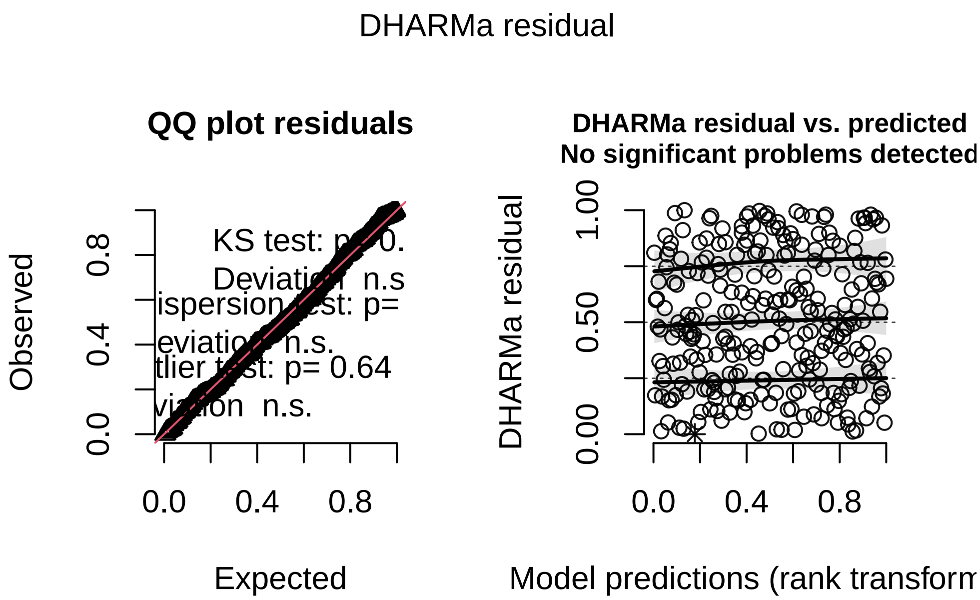
```
# Put models into AIC to see which is the better model to use for the data
AIC(model1, model2)
```

```
       df     AIC
model1  4 20.5207
model2  6 24.5207
```

The best model as determined by Akaike's Information Criterion (AIC) is one that includes tree height and ant species as predictor variables. The model has a lower AIC (20.52) compared to the model that included the interaction between height and ant species (24.52) which had the interaction between tree height and ant species. This implies that the model selected is better in explaining and predicting the data as well without including unnecessary complexity in the visualization.

2g.

```
# Simulate residuals using Dharma package
model1_residual <- simulateResiduals(fittedModel = model1)
plot(model1_residual)
```

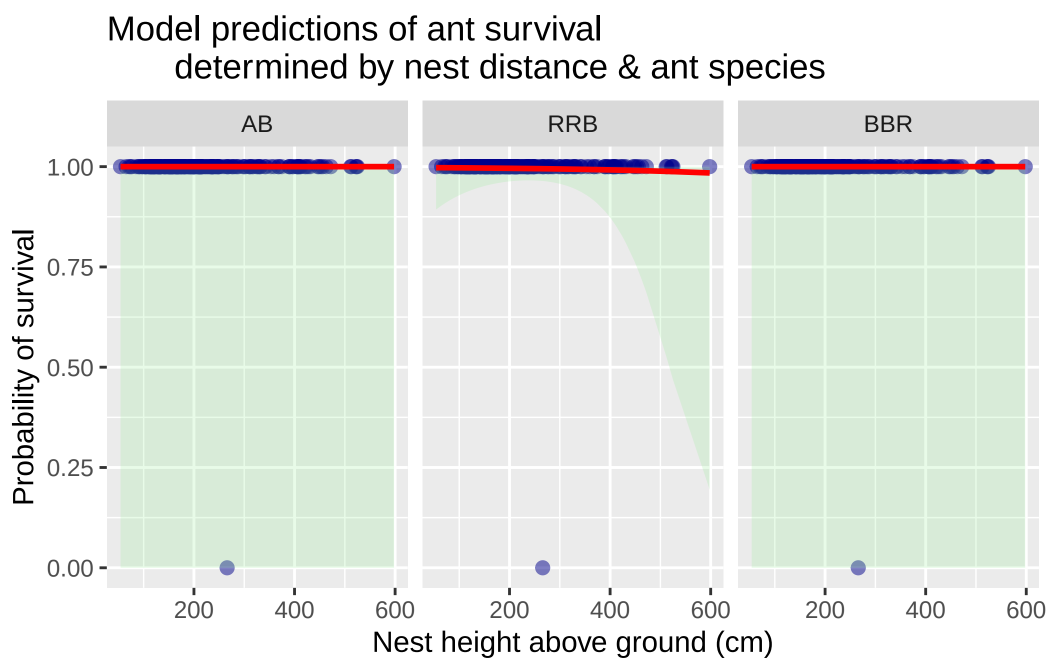


2h.

```
# Generate probabilities / CI
predictions1 <- ggpredict(model1, terms = c("height_cm [all]", "ant_species"))
# Converting data so that it can be plotted on a graph
predictions1 <- as.data.frame(predictions1)

ggplot() +
  # Plot underlying data, input x y axes
  geom_point(data = nests_clean, aes(x = height_cm, y = alive),
             color = "darkblue", alpha = 0.5, size = 2) +
  # Confidence interval, set up in addition to the data visualization w/ ggplot
  geom_ribbon(data = predictions1,
              aes(x = x, ymin = conf.low, ymax = conf.high),
              fill = "lightgreen",
              alpha = 0.2) +
  # Set up probability line across ant species
  geom_line(data = predictions1,
            aes(x = x, y = predicted),
            color = "red", linewidth = 1) +
  # Separate the data plots by ant species
  facet_wrap(~ group) +
  # Set up x y title labels for graph
```

```
  labs(x = "Nest height above ground (cm)",
       y = "Probability of survival", title = "Model predictions of ant survival
       determined by nest distance & ant species")
```



2i.

Figure 1. Model predictions of ant survival across species and nest height. Probabilities of survival were estimated using AIC and linear regression that included nest height above the ground, and various species of ants. The dark blue points represent the amount of ant that survived per species while the ones on the bottom are representing the ones that did not survive. The line represents the model predictions of ant survival. The green shaded regions represent the 95% confidence interval. The results are distributed across each ant species. This information was extracted from the nests_clean dataset.

2j.

```
# Calculate predicted probability at 600 cm
predict2j <- predict(
  model1,
  newdata = data.frame(
    # Split calculated pred. probability across ant species
    height_cm = c(600, 600, 600),
    ant_species = factor(
```

```
      c("RRB", "AB", "BBR"),
      # Generate predictions + distribute them
      levels = levels(nests_clean$ant_species)
    )
  ),
  # Give a probability rather than ratio
  type = "response"
)
predict2j
```

```
        1         2         3 
0.9841796 1.0000000 1.0000000 
```

2k.

I noticed that nest occupancy was higher in taller trees. At 600 cm, the model predicted high likelihood of nest occurrences across the three species of ants, the probability ranging from 0.984 for Mimosae ants to 1 for both Sjostedti and Nigriceps ants. These predictions suggest that height and types of ants are influential variables in nest appearance, and that taller trees have more nest occupancy. This makes sense because taller trees are higher off the ground, away from many predators, and provide more shade as there are more leaves as the nest is higher, providing safety, shelter and shade for the birds that create their nests. Additionally, higher trees are also farther away from any human disturbances. Ant species being an influential factor for nest occupancy likelihood is also relevant. Though the relationship between ants and nesting may be mutualistic, each ant may have different ways of defending nests when they are on the trees, making some habitats more favorable to birds than others.

Problem 3

3a.

My visualizations in Homework 2 were very different because a lot of them were just focused on getting the data on the document, and explaining it conceptually and statistically. Homework 3 however was really different because the data visualizations now had a new goal of engaging the viewers of my visualization. While preserving the data, I had to now find ways on how to creatively display it in ways that are meaningful yet statistically insightful. Across both homework sets they are similar because they both had the same goal of displaying and representing my data informatively and concisely, and used visual patterns and methods to do so. For me specifically, my visualizations across my homework remain consistent because they both make efforts to convey the data in a concise manner. But I think in homework 2, the visualization was more focused on just simply getting the data across through statistical relationships and trends while in homework 3 I got to apply the statistical relationships and trend in a more creative way, that would engage my audience. One set of feedback I received was to include more data about my statistical relationship/trend that I was analyzing. I think this is important and I will definitely implement it because it would make my data more reliable, informative, and accurate.

3b.

Based on my response variable, I think logistical regression is the appropriate test to use to see if height affects nest occurence. It works because the probability of nest occurence is split into two responses, either Yes or No, making it a binary dataset. LR works because it models the possibility of a binary event happening or not in response to a continuous predictor variable, such as tree height. Therefore, it is an appropriate test to see whether tree height really does affect the likelihood of nests appearing.

3c.

```
# Define object and simulate residuals using Dharma package
assumptionchecking <- simulateResiduals(model1)
assumptionchecking
```

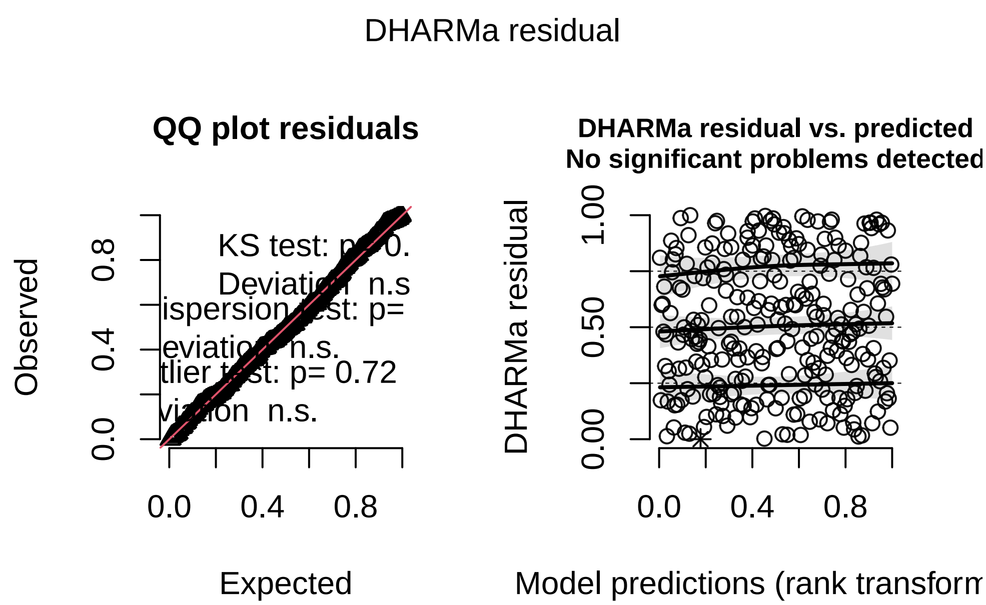
```
Object of Class DHARMa with simulated residuals based on 250 simulations with refit = FALSE

Scaled residual values: 0.6763656 0.4472002 0.4800467 0.7974476 0.8417914 0.7454823 0.817750
```

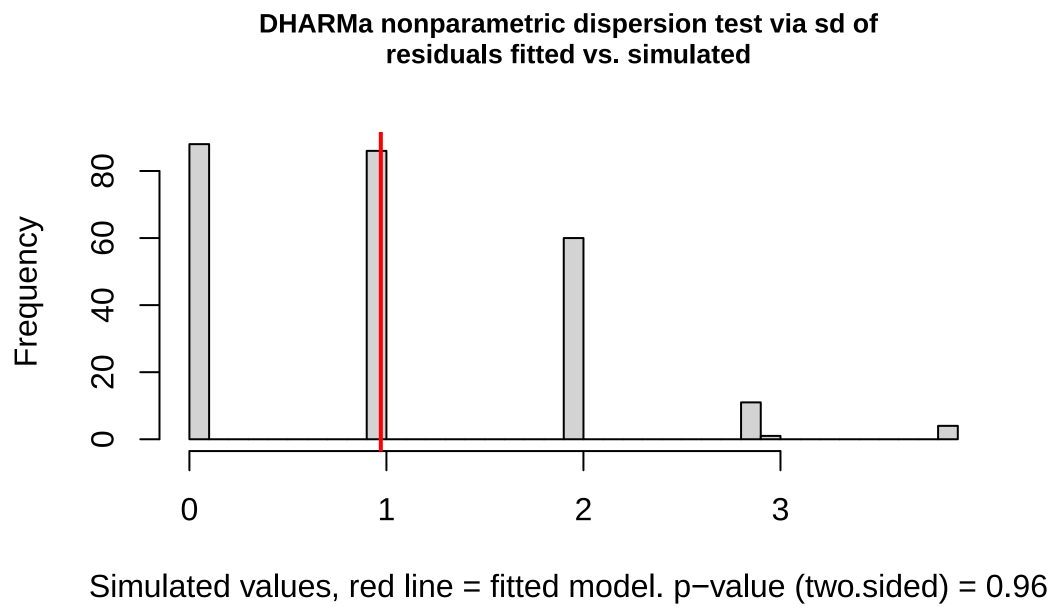
```
# Binary checking
table(nests_clean$alive)
```

```
  0   1
  1 269
```

```
# Diagnostics
plot(assumptionchecking)
```



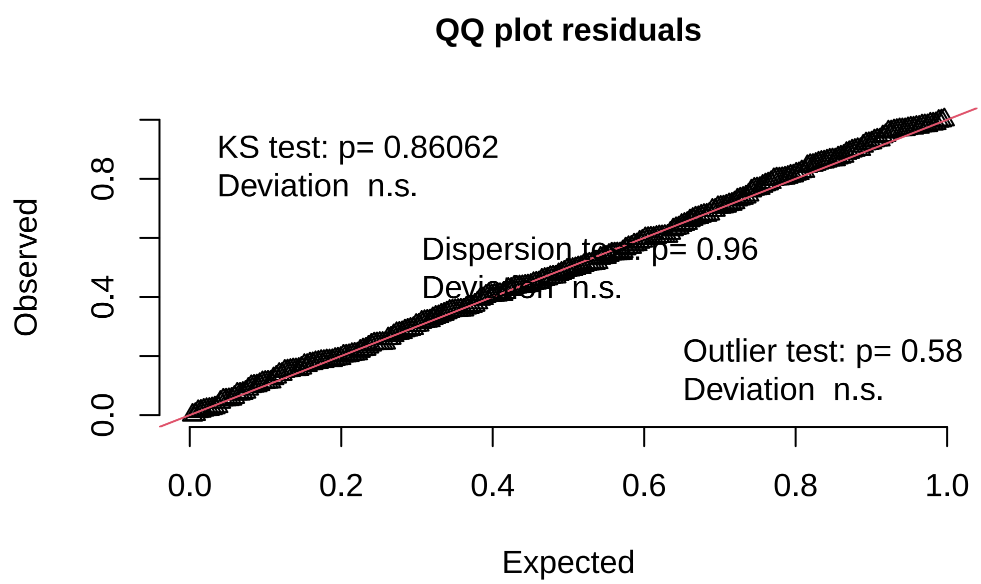

```r
# Testing for overdispersion
testDispersion(assumptionchecking)
```



```
    DHARMa nonparametric dispersion test via sd of residuals fitted vs.
    simulated

data:  simulationOutput
dispersion = 0.97318, p-value = 0.96
alternative hypothesis: two.sided
```

```
# Testing for residuals to be uniformly distributed
testUniformity(assumptionchecking)
```




```
    Asymptotic one-sample Kolmogorov-Smirnov test

data:  simulationOutput$scaledResiduals
D = 0.036682, p-value = 0.8606
alternative hypothesis: two-sided
```

Effect size calculation:

```r
# Define object with relevant variables using logistic regression values
effectsize <- glm(alive ~ height_cm, data = nests_clean, family = binomial)
```

```r
# Calculate effect size
exp(coef(effectsize))
```

```
(Intercept)   height_cm
699.7469268   0.9959759
```

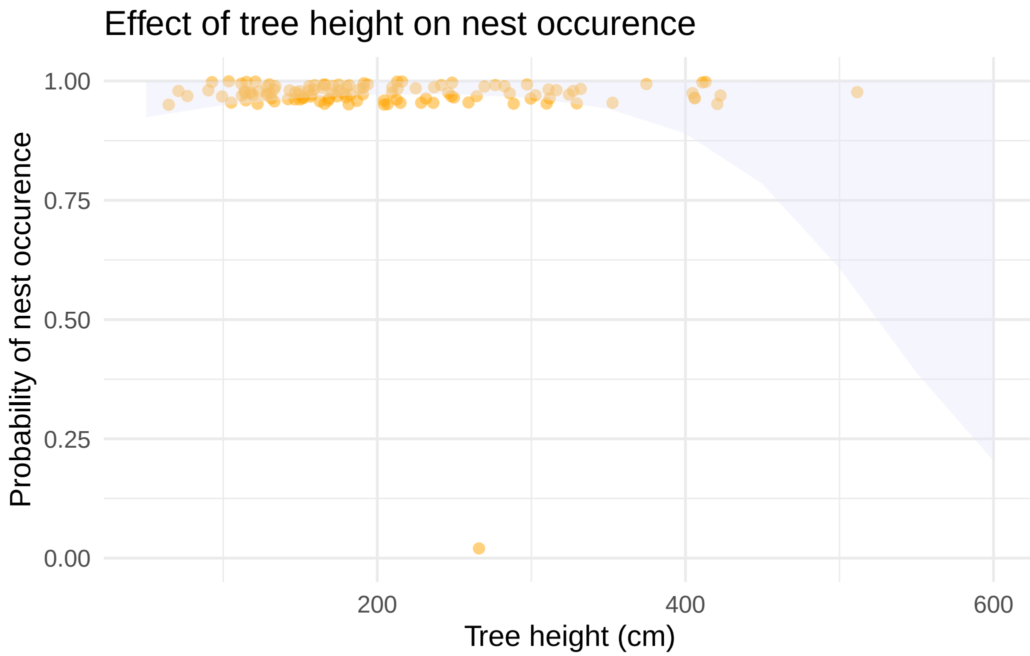
Nest occurence shows to be binary (nest = 1, no nest = 0), satisfying logistic regression use. Simulated residuals from Dharma and assumption checking of logistic regression was used. The tests shown were to check for assumptions and if the data was uniformly distributed and dispersed evenly. An effect size test was ran through defining variables through logistic regression model and then calculating odds ratio, resulting in tree height with a ratio of 0.996, suggesting that for every one cm increase in tree height the probability of nests occurring was multiplied by 0.996. However, because the ratio is very close to 1, that means the ratio is equally distributed, and that tree height actually doesn't have the most significant impact on nest appearances.

3d.

```r
# Define object from logistic regression analysis for model representation
height_3d <- ggpredict(
  effectsize,
  terms = "height_cm"
)
```

Plotting:

```r
ggplot() +
  # Form data plot and model predictions
  geom_jitter(
    data = nests_clean, aes(x = height_cm, y = alive), width = 0.5,
    height = 0.05, alpha = 0.5, color = "orange") +
  # Confidence interval
  geom_ribbon(data = height_3d, aes(x = x, ymin = conf.low, ymax = conf.high),
              fill = "lavender", alpha = 0.4) +
  # Set up x y axes
  labs(x = "Tree height (cm)", y = "Probability of nest occurence",
      title = "Effect of tree height on nest occurence") +
  # Set range for y axis in terms of probability
  scale_y_continuous(limits = c(0, 1)) + theme_minimal()
```



3e.

Figure 1. Tree height's effect on the probability of nest occurrence. The purple shaded part of the graph represents the 95% confidence interval surrounding the logistic regression data, and the area under the curve where the chance of nests appearing is most likely to occur, under trees at 600 cm. The orange points represent individual observations of nests across different trees and different heights where values closer or at 1 represent a nest being present, and values close to or at 0 represent a nest not being there. In summary, nest occurrence remains prominent across most tree heights, but the data seems to be less clustered as the tree height increases as fewer observations are seen.

3f.

Tree height came to have very small influence on the probability of nest occurence, with the odds ratio being close to 1(0.996). A logistic regression test was used to observe the relationship between tree height and probability of nest occurrence. Nests were frequent in the data set ($n = 270$, n = No. of trees), where nests appeared in 269 trees and were absent in one tree (Figure 1). Based on these results it can be suggested that tree height is not a strong predictor to determine probability of nest occurrence.